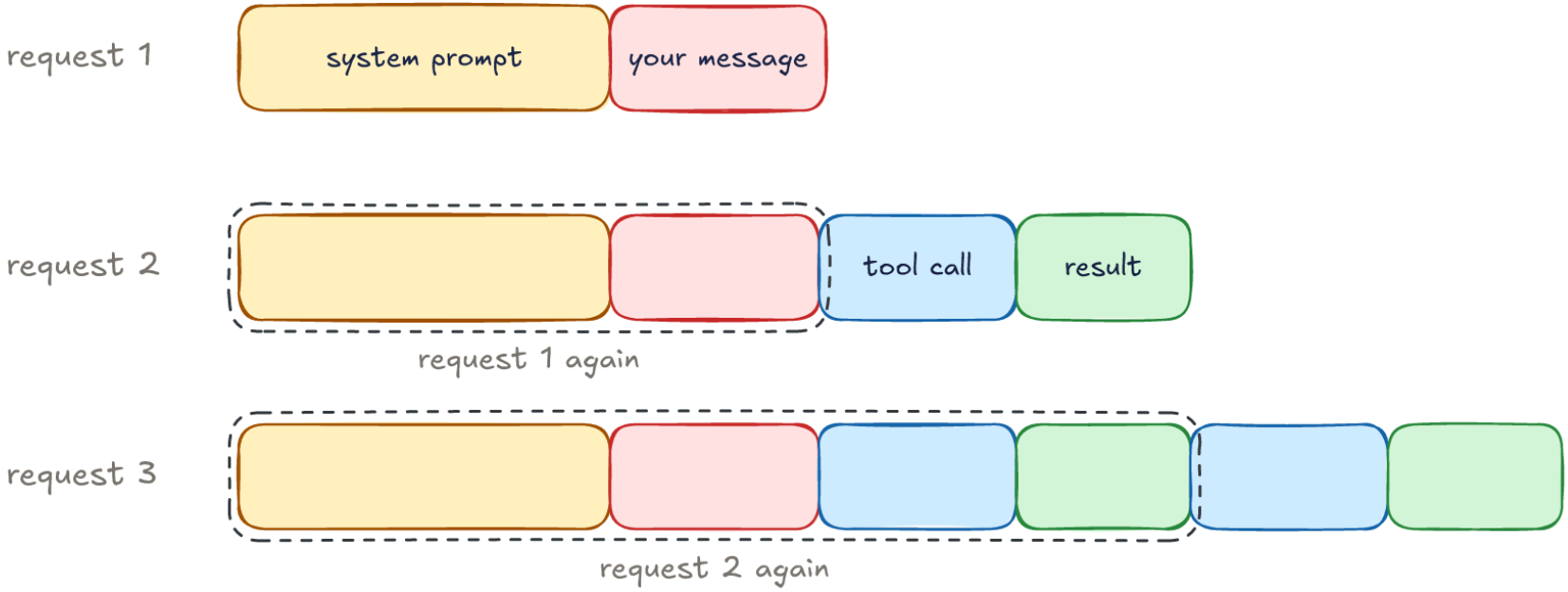
Compaction and Prompt Caching

EECS 498 AASE, Lecture 15

What each request carries



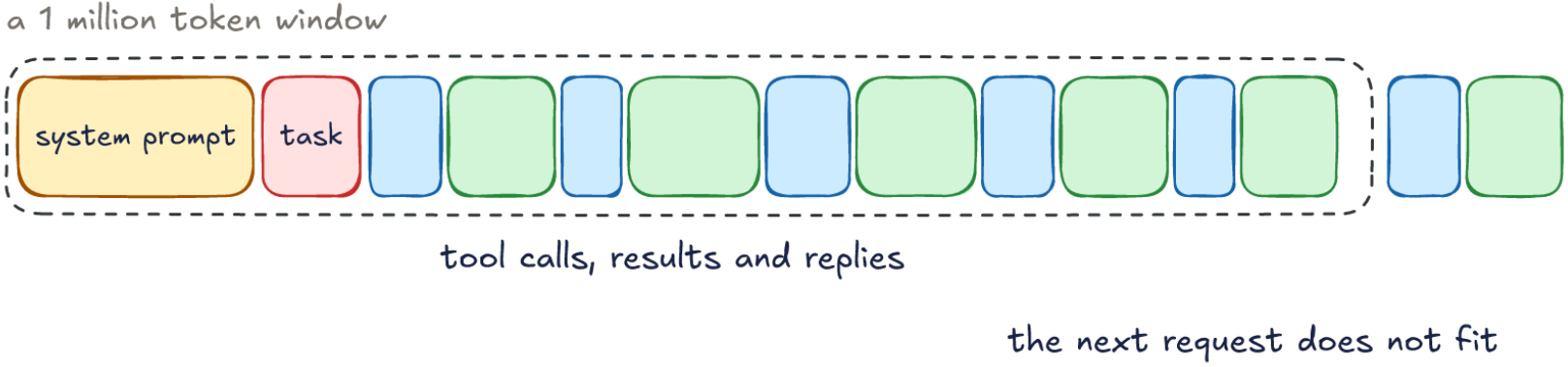
Today's schedule

- [x] Compaction
- [ ] Break
- [x] Prompt caching
- [ ] Questions

Part 1

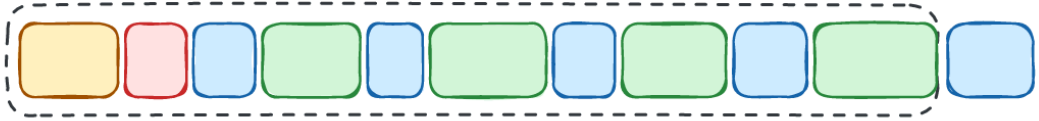
Compaction

A window that fills



Four ways to handle a full window

Stop with an error



Drop the oldest messages



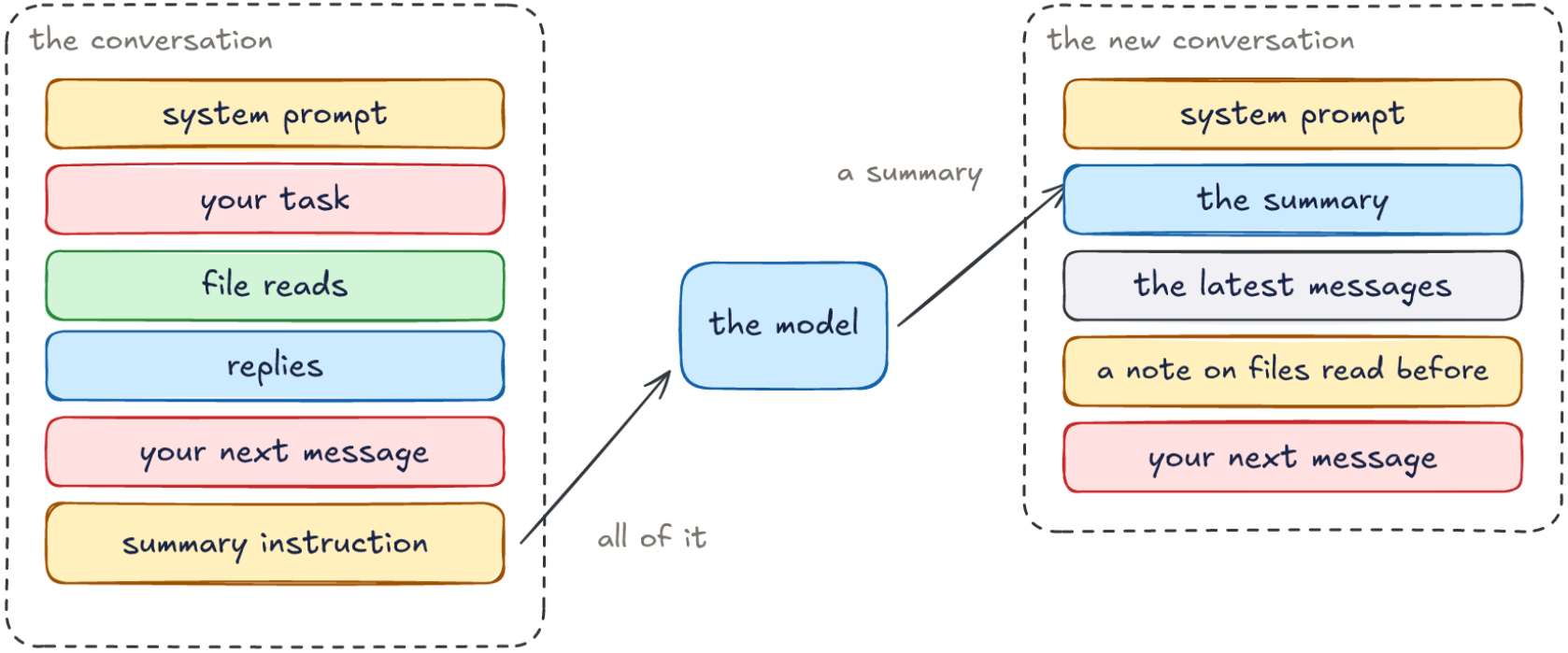
Clear old tool results



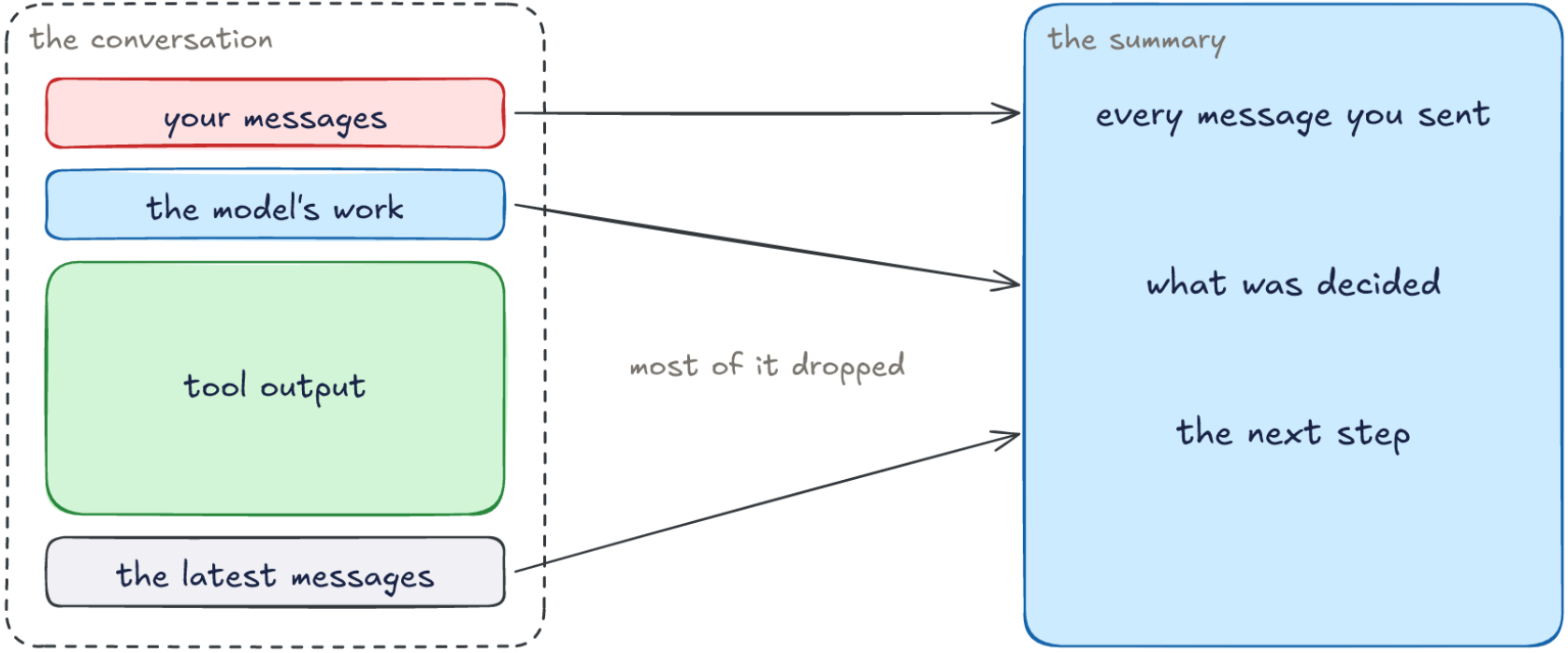
Summarize and start over



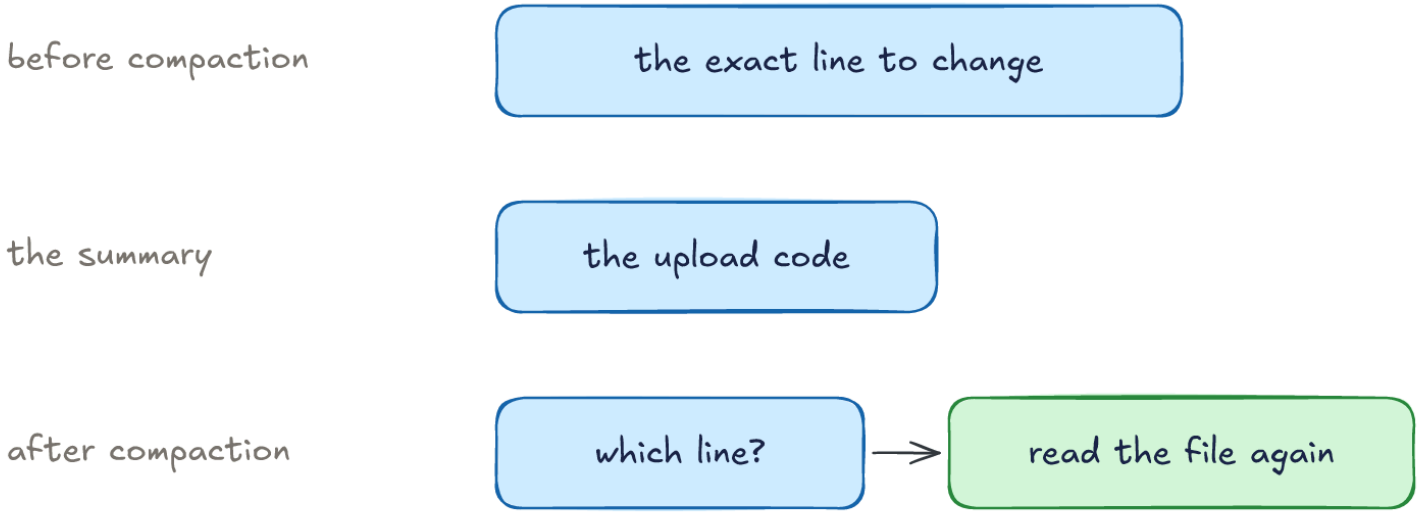
How a compaction runs



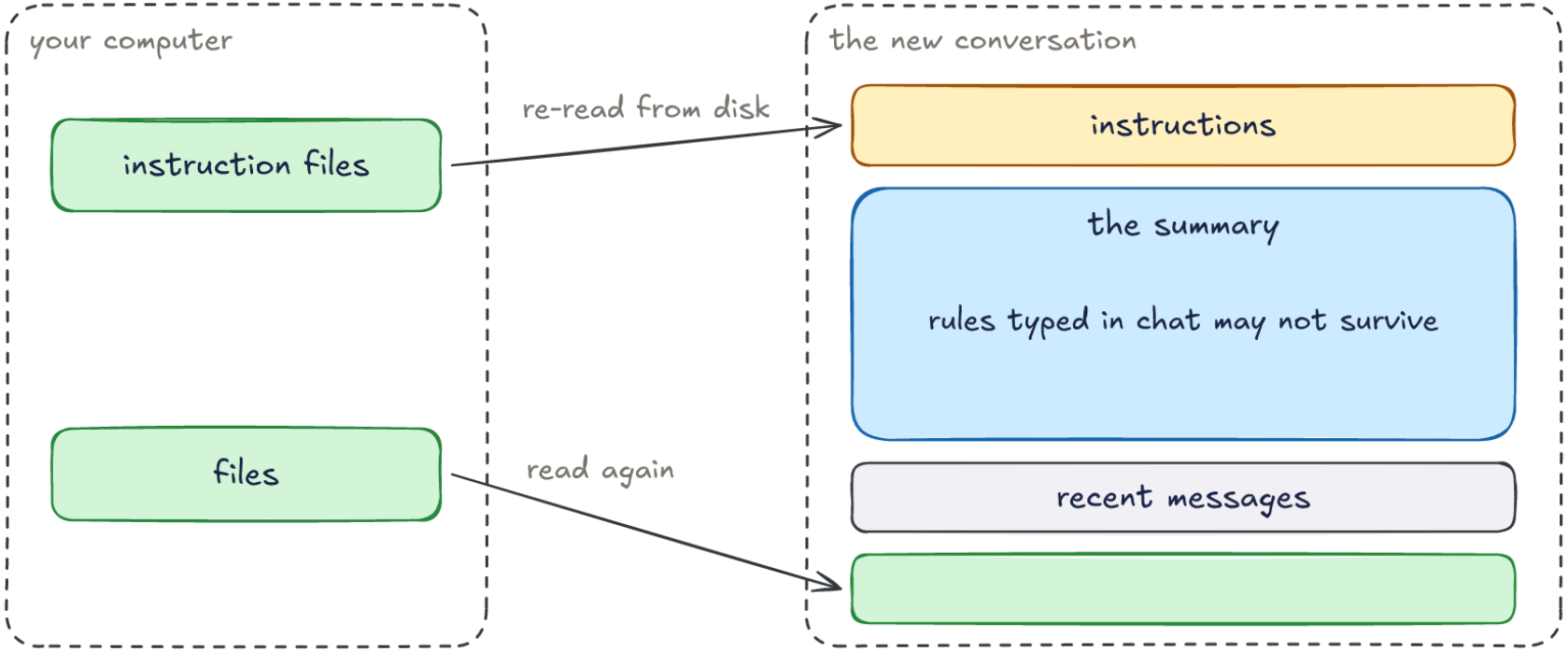
What a summary keeps



What a summary loses



What survives compaction



Compaction as a handoff



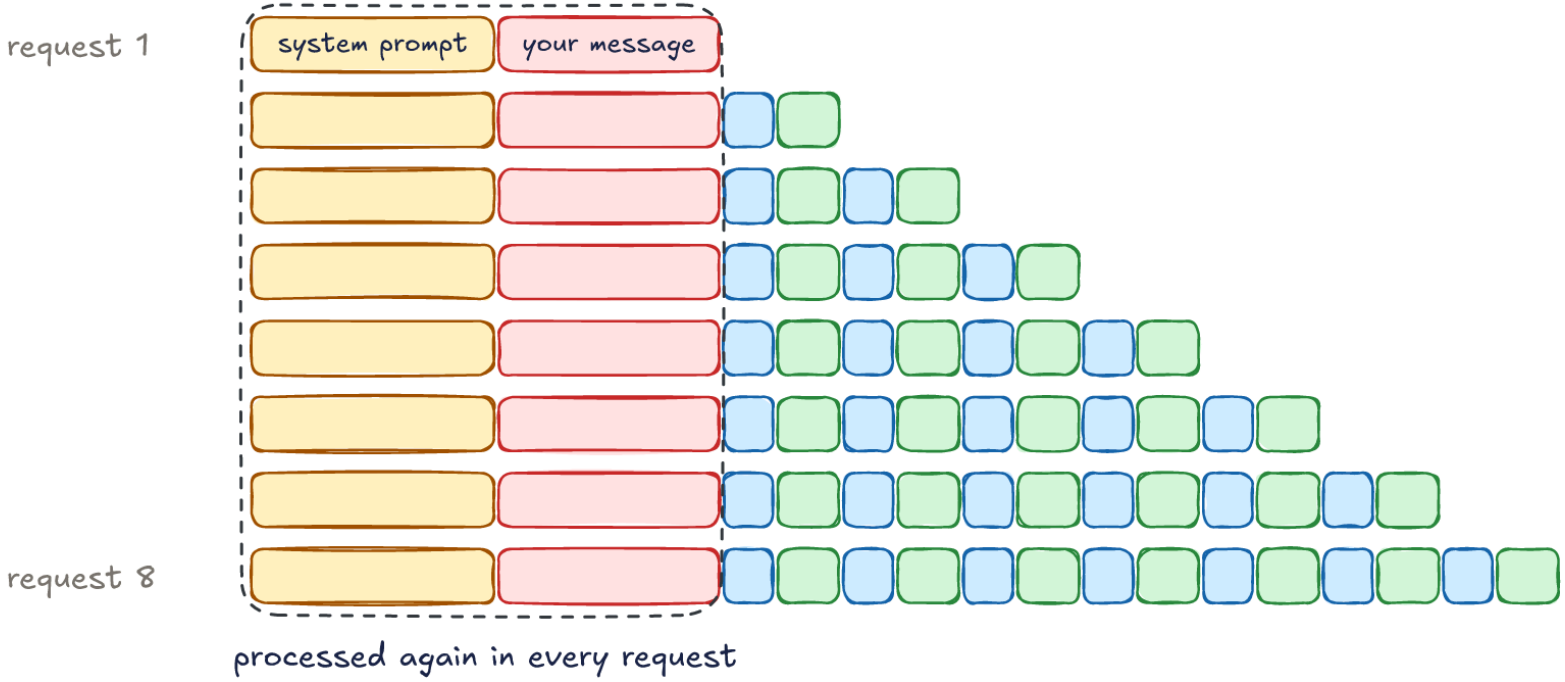
Chat break

Back in five minutes

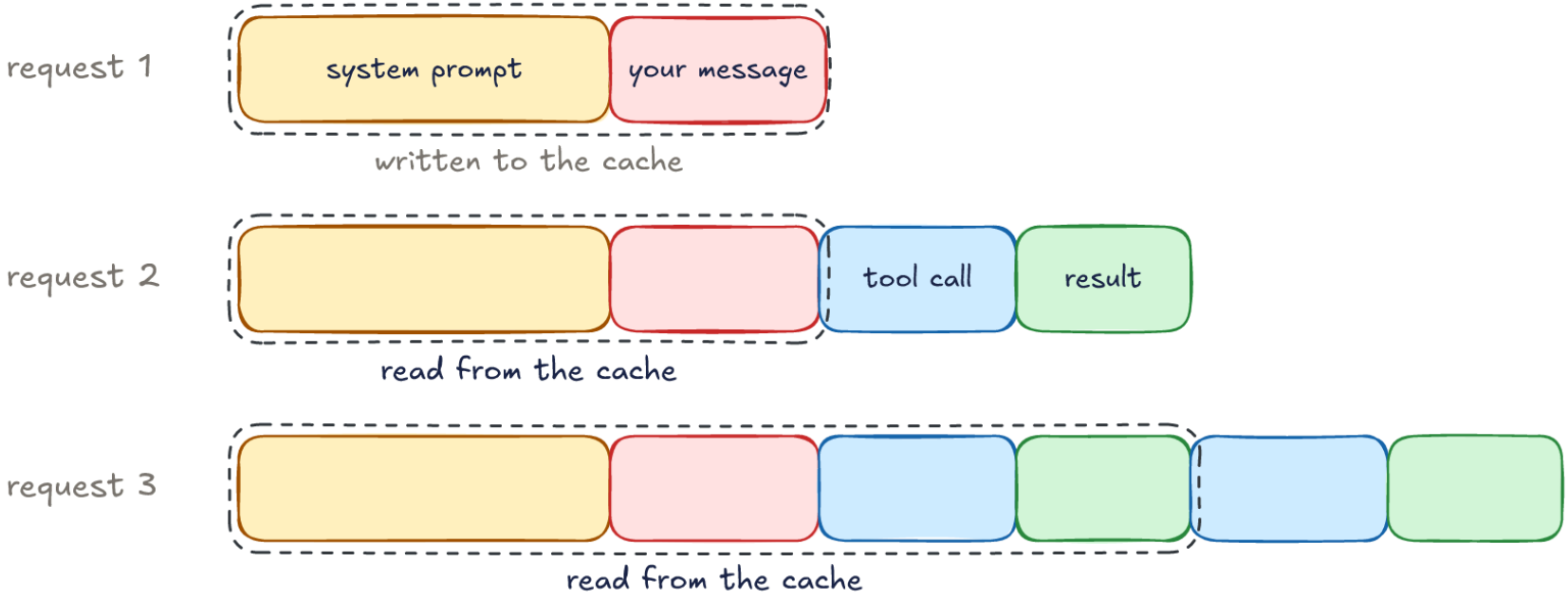
Part 2

Prompt caching

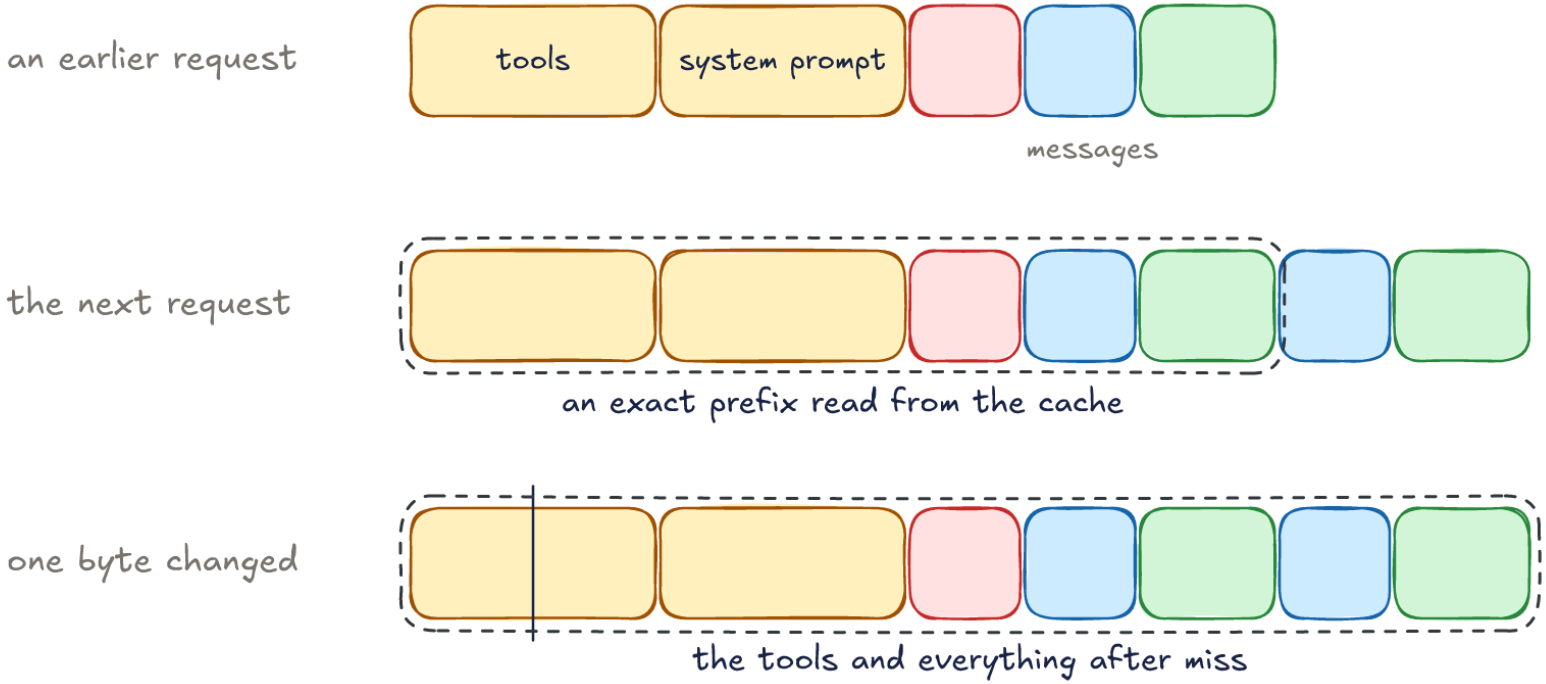
The cost of resending



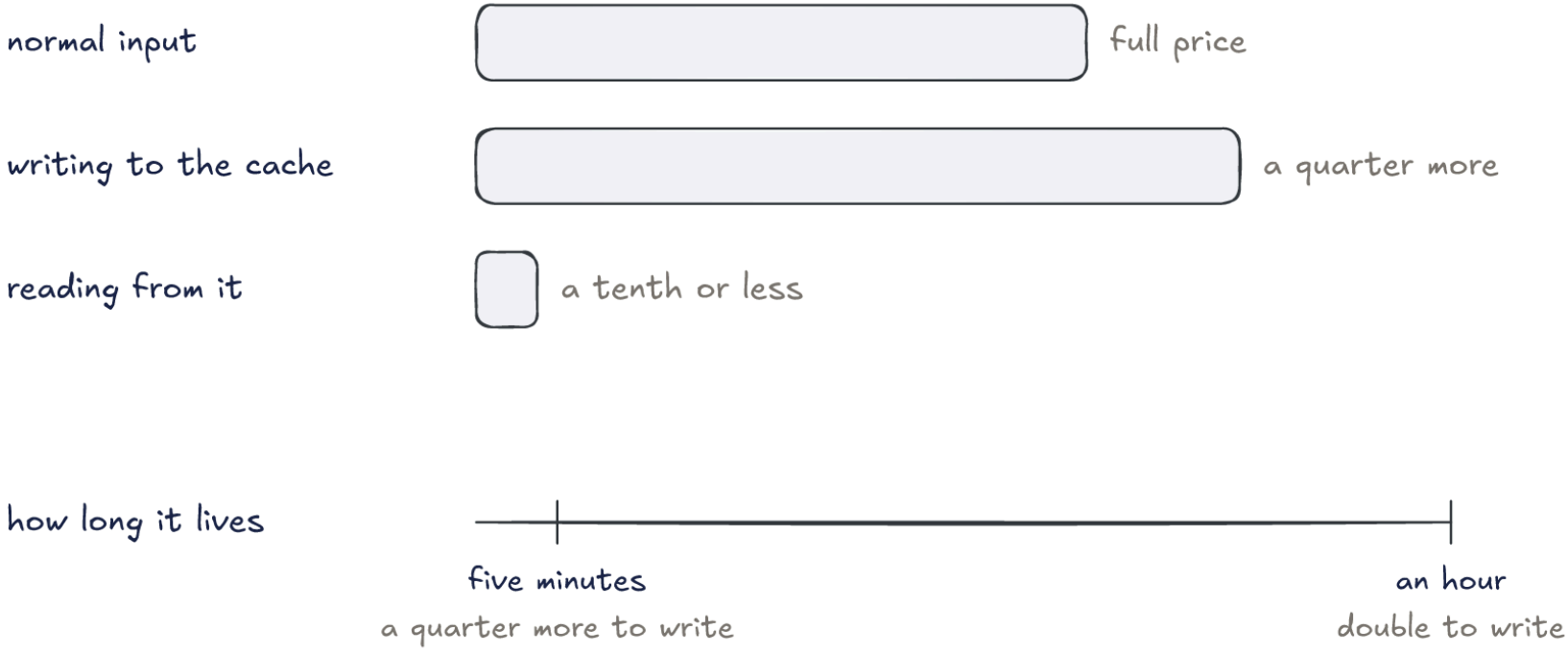
What the cache keeps



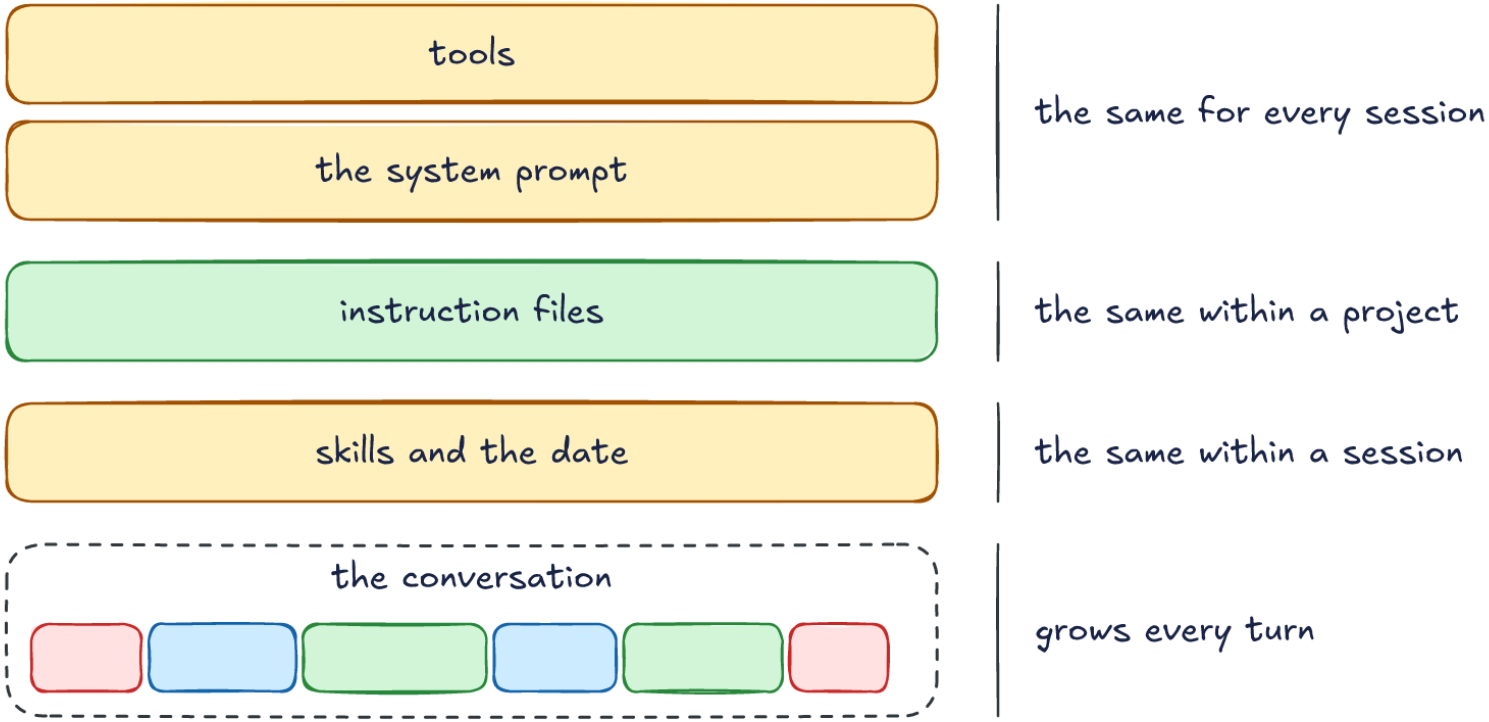
How the cache matches



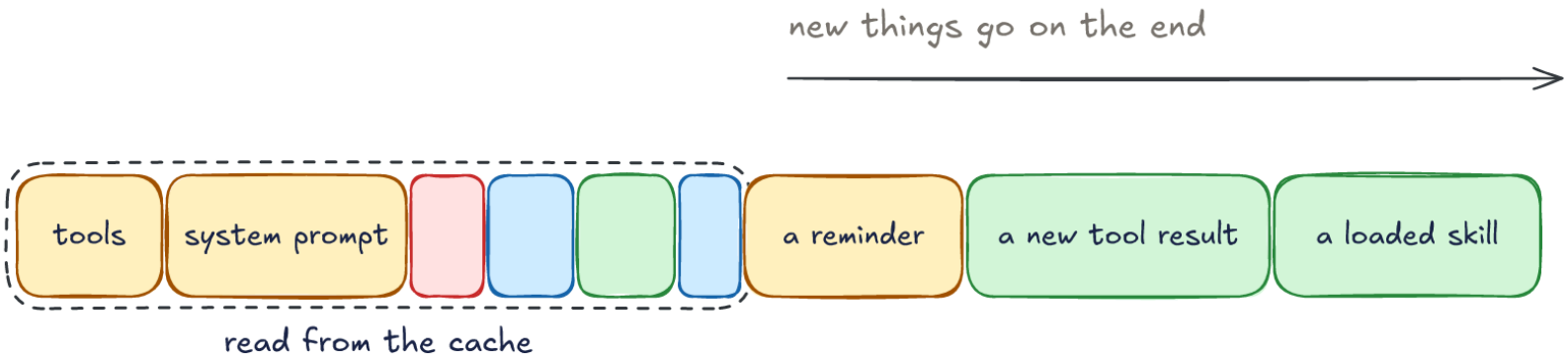
What the cache costs



Request order

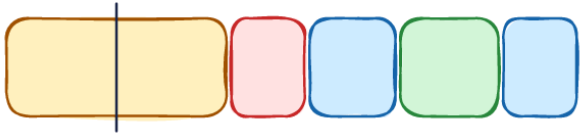


Adding to the end



What resets the cache

A changed tool definition



nothing read

A different model



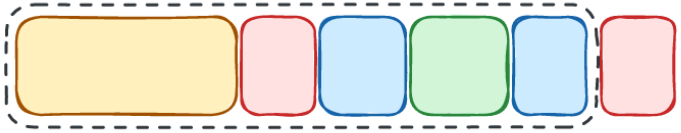
nothing read

A fresh subagent



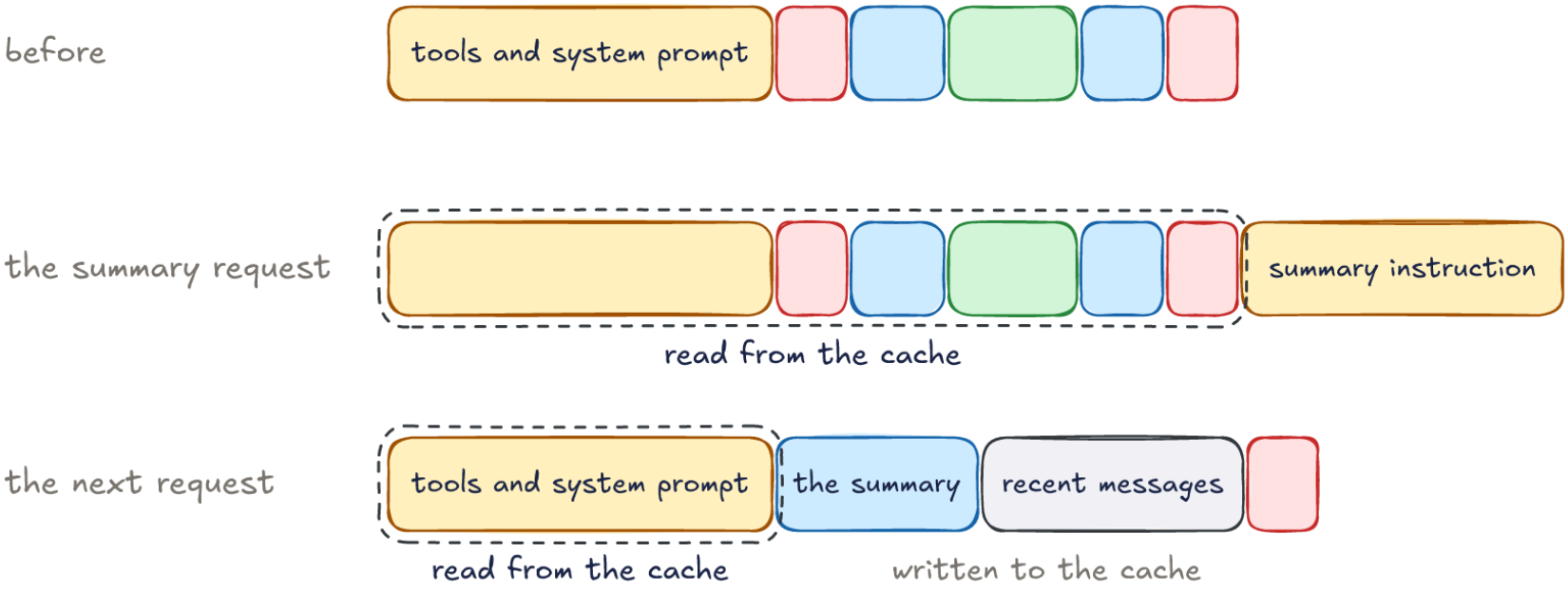
starts cold

A fork



reads the parent's cache

Compaction through the cache



What to take away

- The harness compacts before the window fills
- A summary keeps decisions and loses details
- The cache rewards a harness that only appends
- Every edit to the past is paid again

This week

- Aider summarizes only finished turns
- Each tool round should append

Sources

The Claude Code docs on prompt caching
code.claude.com/docs/en/prompt-caching

The Claude Code docs on what survives compaction
code.claude.com/docs/en/context-window

Lessons from building Claude Code
claude.com/blog/lessons-from-building-claude-code-prompt-caching-is-everything

Anthropic's prompt caching docs
platform.claude.com/docs/en/build-with-claude/prompt-caching

OpenAI's prompt caching docs
developers.openai.com/api/docs/guides/prompt-caching

More sources

The Codex compaction prompt
`github.com/openai/codex`

Aider's chat summarizer
`github.com/Aider-AI/aider`

Anthropic on effective context engineering
`anthropic.com/engineering/effective-context-engineering-for-ai-agents`

Questions

AASE Lecture 15, Compaction and Prompt Caching